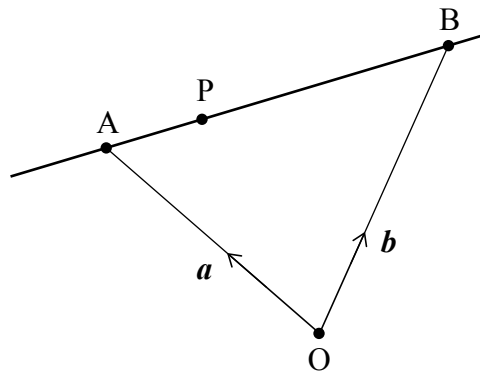


8. [Maximum mark: 8]

The following diagram shows two points A and B such that $\vec{OA} = \mathbf{a}$, $\vec{OB} = \mathbf{b}$.



The point P lies on (AB) so that $\vec{AP} = \lambda \vec{AB}$ where $0 < \lambda < 1$.

(a) Show that $\vec{OP} = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}$. [1]

It is given that $|a| = 1$, $|b| = 2$ and $a \cdot b = \frac{1}{4}$.

(b) In the case that $\vec{OP}$ is perpendicular to $\vec{AB}$, find the value of λ . [7]

[illegible]